Shile Wang

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Education

Peking University (PKU), Beijing, China *09/2023 – Present*
Bachelor of Mathematics and Applied Mathematics, School of Mathematical Sciences

- Major GPA: 89/100 (3.72/4.00)
- Related Courses: Geometry; Differential Geometry; Topology; Differential Manifold; Homology Theory; Geometric Group Theory; Characteristic Classes

Stony Brook University, NY, US *08/2026 – Present*
Exchange Student in Mathematics

- Selected Courses (in progress): Number Theory; Differential Geometry; Complex Geometry

Research Interests

I am interested in low-dimensional topology and geometric group theory, in particular mapping class groups, CAT(0) spaces, and group actions on specific manifolds.

Research Experience

Mapping Class Group *09/2025 – Present*
Instructor: Prof. Yi Liu (Beijing International Center for Mathematical Research)

- Studied B. Farb and D. Margalit's *A Primer on Mapping Class Groups* and read relevant papers.
- Learned about mapping class groups via the Birman exact sequence, the symplectic representation, and the Johnson filtration, and studied basic properties of Teichmüller space. Also learned about homomorphisms between mapping class groups, braid groups, and transformation groups.
- Studied the action of mapping class groups on high-dimensional spheres, proved that when $g \geq 3$, there exists a faithful action of $\mathrm{Mod}(S_g)$ on the $(2g-1)$-sphere, and authored a preprint on arXiv: *arXiv:2610.10577*.

Fixed Point Properties of Group Actions on CAT(0) Spaces *06/2026 – 08/2026*
University of Chicago Mathematics REU
Mentor: Zhong Zhang

- Studied J.-P. Serre's *Trees*, B. Farb's *Group Actions and Helly's Theorem*, and M. R. Bridson's *Semisimple Actions of Mapping Class Groups on CAT(0) Spaces*.
- Learned about how to prove property FA_n for reflection groups, arithmetic groups, mapping class groups, and braid groups.
- Wrote an expository article: *Fixed Point Properties of Group Actions on CAT(0) Spaces*.
- Gave a talk on fixed point properties of group actions on trees.

CAT(0) Cube Complexes *03/2026 – 06/2026*
Peking University Low-Dimensional Topology Seminar
Instructors: Prof. Yi Liu, Prof. Jiajun Wang, Prof. Wenyuan Yang

- Studied P. Schwer's *CAT(0) Cube Complexes: An Introduction*, and learned the basic properties of CAT(0) cube complexes and studied cubulations of half-space systems.
- Presented part of J. Crisp and B. Wiest's *Quasi-isometrically embedded subgroups of braid and diffeomorphism groups*, focusing on how to embed right-angled Artin groups into mapping class groups.

Knots and Links *10/2025 – 12/2025*
Peking University Low-Dimensional Topology Seminar
Instructors: Prof. Yi Liu and Prof. Jiajun Wang

- Studied D. Rolfsen's *Knots and Links* and presented part of Chapter 3.
- Learned the Wirtinger presentation and used it to compute the fundamental groups of knots and links and to study the properties of horned spheres. Also learned the basic techniques to study knots and links, such as Seifert surfaces and Alexander invariants.

CAT(0) Spaces *09/2025 – 12/2025*
Peking University Geometric Group Theory Seminar
Instructor: Prof. Wenyuan Yang

- Studied Sections 2.1–2.8 of M. R. Bridson and A. Haefliger's *Metric Spaces of Non-Positive Curvature* and presented Section 2.6.
- Learned the basic properties of CAT(0) spaces. Studied the general structure of groups of isometries of CAT(0) spaces; also became familiar with cone topology, horofunctions, and Busemann functions.

Heegaard Splitting *03/2025 – 06/2025*
Peking University Low-Dimensional Topology Seminar
Instructors: Prof. Yi Liu and Prof. Jiajun Wang

- Studied M. Scharlemann's *Heegaard splittings of compact 3-manifolds* and presented Section 2.
- Learned the definition of Heegaard splitting and Heegaard diagram, and how to read off basic algebraic topological information from a Heegaard diagram.
- Became familiar with basic tools in low dimensional topology such as handle decomposition and Morse theory.

Geometry of Surfaces *02/2025 – 06/2025*
Peking University Geometry Seminar
Instructor: Assoc. Prof. Zhiqiang Bao

- Studied J. Stillwell's *Geometry of Surfaces* and presented Sections 5.6–6.10.
- Learned the classification of Euclidean, spherical, and hyperbolic surfaces (Killing-Hopf theorem) by studying the groups of isometries of surfaces.
- Studied the classification of closed geodesic paths through the fundamental groups of surfaces.
- Learned basic results in tessellation such as the properties of fundamental regions and orbifolds.

Honors & Awards

- Academic Excellence Award, Peking University 2026
- Mo Li Fang Scholarship 2026
- Yi Zheng Scholarship 2024
- Second Prize, Chinese College Student Mathematical Contest in Beijing area 2023 & 2024
- Gold Medal, Chinese Mathematical Olympiad (CMO) 2022
- Gold Medal, Chinese Girls' Mathematical Olympiad (CGMO) 2022

Other Information

- Lifelong passion and solid background for mathematics: participated in Mathematical Olympiad since primary school and won many awards.
- Standardized tests: GRE Mathematics Subject Test: 970 (95%).
- Language skills: fluent in English (TOEFL: 103).
- Strengths and interests: enjoy swimming (won three silver medals and one bronze medal at PKU).